# Design Proposal: Mach–Zehnder Interferometers

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**Abstract**

This paper presents the design, fabrication, simulation, and experimental characterization of silicon strip waveguide Mach–Zehnder interferometers (MZIs) for group-index extraction on a 220 nm SOI platform. Five quasi-TE MZI devices with 500 nm-wide strip waveguides were implemented using Y-branch splitters/combiners and path length differences of 30 µm, 50 µm, and 100 µm, including two duplicate 50 µm designs for process-variation analysis. A compact waveguide model and analytical MZI transfer function were used to predict transmission spectra and free spectral range (FSR). Measurements from 1.50 µm to 1.58 µm yielded extracted group indices ranging from 4.1103 to 4.1576, in good agreement with the simulated value of 4.197. The measured FSR values also closely matched the calculated results, confirming the validity of the proposed MZI-based group-index extraction method.

## I. Introduction

Silicon photonics has become a leading platform for integrated optics because of high index contrast, compact device footprints, and compatibility with CMOS fabrication processes [1], [4]. By integrating waveguides, splitters, couplers and resonant structures on a single chip, it is possible to implement complex optical signal-processing functions for communications, sensing and computing.

Interferometric circuits such as Mach–Zehnder interferometers (MZIs) are key building blocks in many of these systems. They are widely used for modulation, filtering and sensing, and their operation relies on the coherent interference of optical fields that have accumulated different phases along distinct paths [2], [3].

In this design project, several unbalanced Mach–Zehnder interferometers (MZIs) were designed based on 220 nm-thick silicon strip waveguides to enable extraction of the waveguide group index from the measured free spectral range (FSR). All circuits use a 500 nm-wide waveguide and quasi-TE polarization; the path length difference $\Delta L$ between the arms is varied, and one design is duplicated in two locations to quantify fabrication and process variability.

## II. Theory

In this section, the basic theory and compact equations used in the design and simulation of Mach–Zehnder interferometers (MZIs) are summarized. The goal is not to provide detailed derivations, but rather to define the key building blocks and formulae that will be used later for modeling, simulation, and data analysis. The notation and concepts follow standard photonics texts such as [1], [2], [3].

### A. Effective and Group Index

For a guided mode with effective index $n_{\text{eff}}(\lambda)$, the propagation constant is

$$\beta(\lambda) = \frac{2\pi}{\lambda} n_{\text{eff}}(\lambda). \tag{1}$$

The group index is defined as

$$n_g(\lambda) = n_{\text{eff}}(\lambda) - \lambda \frac{dn_{\text{eff}}}{d\lambda}. \tag{2}$$

To obtain a compact model for use in circuit simulations, we approximate $n_{\text{eff}}(\lambda)$ near a reference wavelength $\lambda_0$ as a low-order polynomial

$$n_{\text{eff}}(\lambda) \approx a_0 + a_1(\lambda - \lambda_0) + a_2(\lambda - \lambda_0)^2, \tag{3}$$

edX public username: `pvhedx`.

where $a_0$, $a_1$, and $a_2$ are obtained by least-squares fitting to mode solver data. Differentiating (3) and substituting into (2) gives

$$\frac{dn_{\text{eff}}}{d\lambda} \approx a_1 + 2a_2(\lambda - \lambda_0), \tag{4}$$

$$n_g(\lambda) \approx n_{\text{eff}}(\lambda) - \lambda\left[a_1 + 2a_2(\lambda - \lambda_0)\right]. \tag{5}$$

These expressions provide a compact model for both $n_{\text{eff}}$ and $n_g$ suitable for use in MATLAB or Lumerical INTERCONNECT.

### *B. MZI Transfer Function*

We consider an unbalanced MZI formed by two nominally identical 50/50 Y-branch splitters and two arms of length $L_1$ and $L_2 = L_1 + \Delta L$. Neglecting loss imbalance, the phase difference between the two arms is

$$\Delta\phi(\lambda) = \beta(\lambda)\Delta L = \frac{2\pi}{\lambda} n_{\text{eff}}(\lambda)\Delta L. \tag{6}$$

For ideal 3 dB splitters, the normalized intensity transmission at one output port is

$$T(\lambda) = \frac{1}{2}\left[1 + \cos\left(\Delta\phi(\lambda)\right)\right]. \tag{7}$$

In practice, finite extinction ratio and splitter imbalance can be captured by introducing a visibility factor $0 \le V \le 1$ and a static phase offset $\phi_0$,

$$T(\lambda) = T_0\left[1 + V\cos\left(\Delta\phi(\lambda) + \phi_0\right)\right], \tag{8}$$

where $T_0$ accounts for insertion loss. In the present proposal, we use (7) as the ideal transfer function for analytical simulations and will refine the visibility and offset parameters after measurements.

### *C. Group Index from Free Spectral Range*

Constructive interference occurs when

$$\Delta\phi(\lambda_m) = \beta(\lambda_m)\Delta L = 2\pi m, \tag{9}$$

for integer $m$. The next maximum at $\lambda_{m+1} = \lambda_m + \Delta\lambda_{\text{FSR}}$ satisfies

$$\Delta\phi(\lambda_{m+1}) = \beta(\lambda_{m+1})\Delta L = 2\pi(m+1). \tag{10}$$

Subtracting and assuming $\Delta\lambda_{\text{FSR}}$ is small,

$$\left[\beta(\lambda_{m+1}) - \beta(\lambda_m)\right]\Delta L \approx 2\pi. \tag{11}$$

Using a first-order Taylor expansion,

$$\left.\frac{d\beta}{d\lambda}\right|_{\lambda_0} \Delta\lambda_{\text{FSR}}\,\Delta L \approx 2\pi. \tag{12}$$

From dispersion theory,

$$n_g = c\frac{d\beta}{d\omega}, \tag{13}$$

and, after transforming from frequency to wavelength and rearranging (12), we obtain the familiar wavelength-domain expression

$$\Delta\lambda_{\text{FSR}} \approx \frac{\lambda_0^2}{n_g(\lambda_0)\,\Delta L}, \tag{14}$$

or equivalently

$$n_g(\lambda_0) \approx \frac{\lambda_0^2}{\Delta\lambda_{\text{FSR}}\,\Delta L}. \tag{15}$$

In this work, the equation (14) was used to extract the FSR values from simulated waveguide group index. Experimentally, the FSR will be obtained by measuring the wavelength spacing between adjacent maxima (or minima) in the transmission spectrum and compared with computed values.

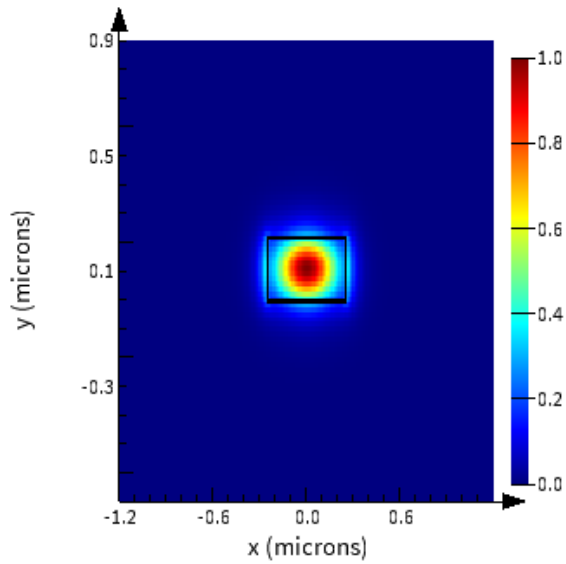


Fig. 1. Simulated fundamental TE mode profile for a $500\,\mathrm{nm} \times 220\,\mathrm{nm}$ silicon strip waveguide at $1550\,\mathrm{nm}$.

## III. Modelling and Simulation

### A. Waveguide Geometry and Mode Profiles

All designs are implemented on a silicon-on-insulator platform with a silicon device layer thickness of 220 nm and a buried oxide cladding. The waveguide geometry is fixed to a single-mode strip with width $W = 500\,\mathrm{nm}$, optimized for the fundamental quasi-TE mode near $\lambda_0 = 1550\,\mathrm{nm}$. Fibre grating couplers and other passive components will be selected from the provided PDK for TE polarization.

The fundamental TE mode profile for the 500 nm $\times$ 220 nm waveguide was simulated using Lumerical MODE Solutions and the resulting electric field magnitude plot is shown in Fig. 1.

### B. Effective and Group Index vs. Wavelength

For the 500 nm-wide waveguide, the effective index $n_{\text{eff}}(\lambda)$ were computed over a wavelength range of 1500–$1600\,\mathrm{nm}$ using the mode solver. The resulting $n_{\text{eff}}(\lambda)$ curve were fitted to the polynomial model in (3), and the corresponding group index $n_g(\lambda)$ were obtained using (2).

Figure 2 shows the simulated group index as a function of wavelength for the TE strip. Figure 3 shows the simulated effective index as a function of wavelength for the TE strip, together with the polynomial fit. The obtained fit model by running script was

$$n_{\text{eff}}(\lambda) \approx 2.443 - 1.131(\lambda - 1.55) - 0.0426(\lambda - 1.55)^2, \tag{16}$$

### C. Interferometer Designs and Parameter Variations

The proposal focuses on five MZI designs, all using the same 500 nm-wide TE strip waveguide and Y-branch splitters. The only varying parameter is the path length difference $\Delta L$; one design is duplicated to study fabrication variability. Table I summarizes the interferometer designs and calucalted FSR at $1550\,\mathrm{nm}$ with $n_g = 4.197$ calculated using (15)

### D. Transmission Spectra

The transmission spectra of the interferometers were computed using the analytical transfer function in (7) together with the compact model for $n_{\text{eff}}(\lambda)$, and cross-checked using circuit-level simulations in Lumerical INTERCONNECT.

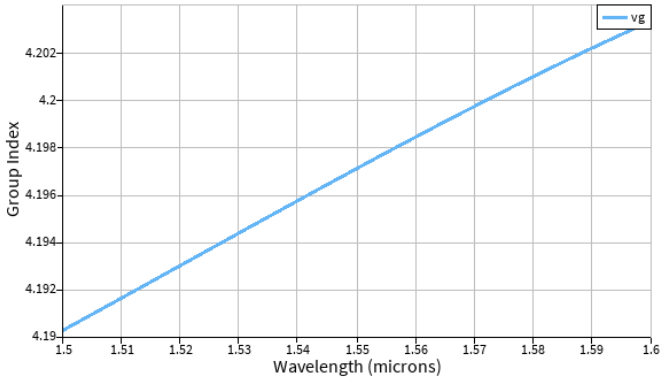


Fig. 2. Simulated group index $n_g$ versus wavelength for the 500 nm-wide TE strip waveguide

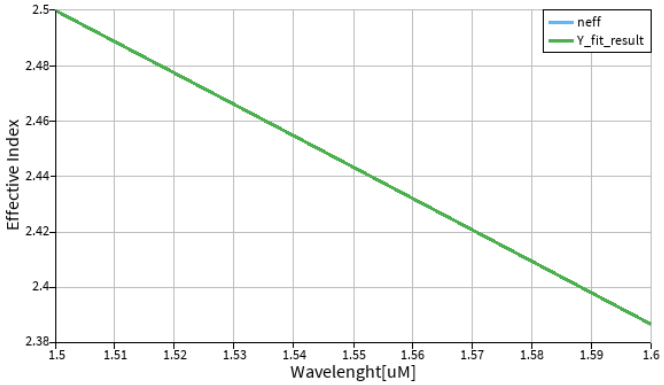


Fig. 3. Simulated effective index $n_{\text{eff}}$ and group index $n_g$ versus wavelength for the 500 nm-wide TE strip waveguide, along with a polynomial compact model fit.

TABLE I
MZI DESIGN VARIANTS AND EXPECTED FREE SPECTRAL RANGE AT $\lambda_0 = 1550\,\text{nm}$, WITH $n_g(\lambda_0) = 4.197$.

| Design ID | $\Delta L$ ($\mu$m) | Polarization | Est. FSR (nm) |
|---|---|---|---|
| MZI_TE_W500_DL30 | 30 | TE | 19.08 |
| MZI_TE_W500_DL50 | 50 | TE | 11.45 |
| MZI_TE_W500_DL100 | 100 | TE | 5.72 |
| MZI_TE_W500_DL50_A | 50 | TE | 11.45 |
| MZI_TE_W500_DL50_B | 50 | TE | 11.45 |

Figure 4 shows a simulated spectrum for MZI_TE_W500_DL30, illustrating a relatively large FSR across the available wavelength range. Figure 5 and Figure 6 shows the spectrum for MZI_TE_W500_DL50 and MZI_TE_W500_DL100 respectively, demonstrating the reduction in FSR as $\Delta L$ increases.

The FSR from these simulated spectra were used with (15) to obtain the group index.

### *E. Monte Carlo Simulation of the* $50\,\mu\text{m}$ *MZI Design*

In addition to the nominal analytical and circuit-level simulations, a Monte Carlo simulation was performed in Ansys for the $50\,\mu\text{m}$ imbalance design (MZ2), since this design was repeated three times on the fabricated chip and was intended for process-variation analysis. The goal of this simulation was to estimate the expected spread in spectral response due to fabrication-induced variations and to compare the measured duplicated devices against the simulated statistical distribution.

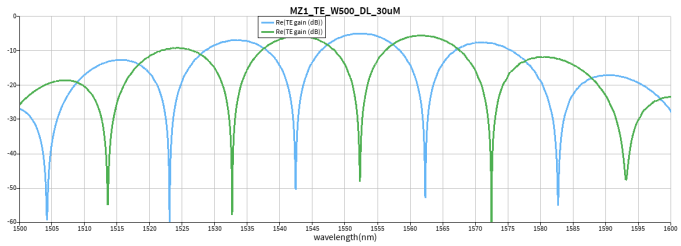


Fig. 4. Simulated transmission spectrum for MZI_TE_W500_DL30, showing a relatively large free spectral range.

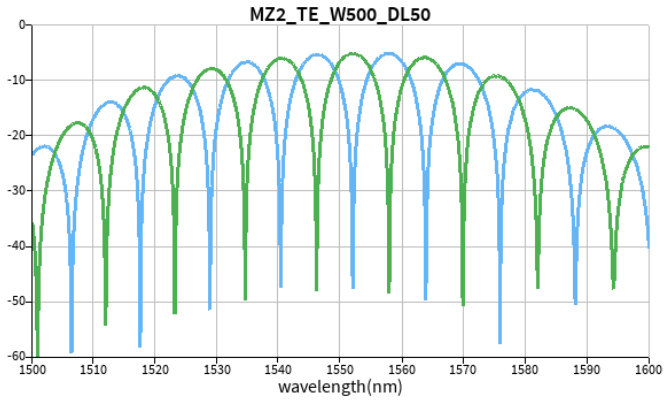


Fig. 5. Simulated transmission spectrum for MZI_TE_W500_DL50, showing a reduced free spectral range due to the larger path length difference.

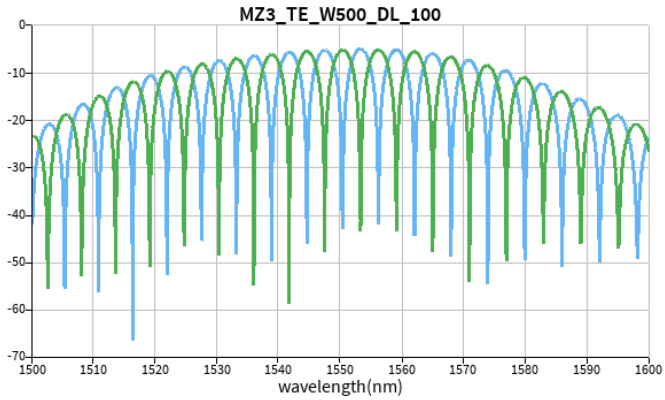


Fig. 6. Simulated transmission spectrum for MZI_TE_W500_DL100, showing a further reduced free spectral range due to the larger path length difference.

Figure 7 shows the Monte Carlo transmission spectra for the MZ2 design. The overlaid responses show the expected variation in resonance positions and fringe spacing across the simulated process distribution. From these simulations, the free spectral range (FSR) was extracted for each trial, and the resulting statistical distribution is shown in Fig. 8. The Monte Carlo simulation produced a mean FSR of $11.5088\,\mathrm{nm}$ with a standard deviation of $0.282\,958\,\mathrm{nm}$. The corresponding distribution of extracted center wavelength is shown in Fig. 9, with a mean wavelength of $1548.79\,\mathrm{nm}$ and a standard deviation of $3.457\,43\,\mathrm{nm}$.

These Monte Carlo results provide an expected process-variation envelope for the $50\,\mu\mathrm{m}$ MZI design. The measured FSR values of the three fabricated $50\,\mu\mathrm{m}$ devices can therefore be compared directly against this simulated distribution to determine whether the fabricated devices fall within the expected process window.

The proposed interferometer designs were fabricated on a silicon-on-insulator (SOI) platform with a $220\,\mathrm{nm}$ silicon device layer. All devices used quasi-TE strip waveguides with a nominal width of $500\,\mathrm{nm}$. The photonic

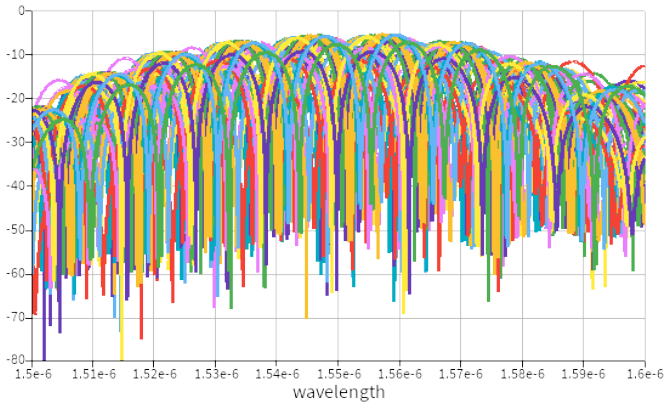


Fig. 7. Monte Carlo simulated transmission spectra for the 50 µm MZI design (MZ2) generated using Ansys. The spread in resonance positions and fringe spacing reflects the expected process-induced variation.

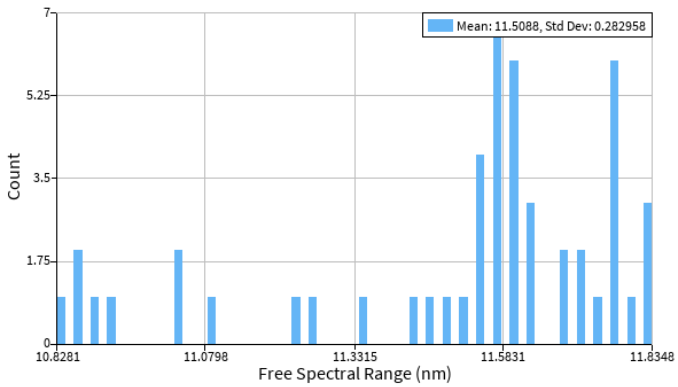


Fig. 8. Histogram of extracted free spectral range from the Monte Carlo simulation of the 50 µm MZI design. The simulated mean FSR is 11.5088 nm with a standard deviation of 0.282 958 nm.

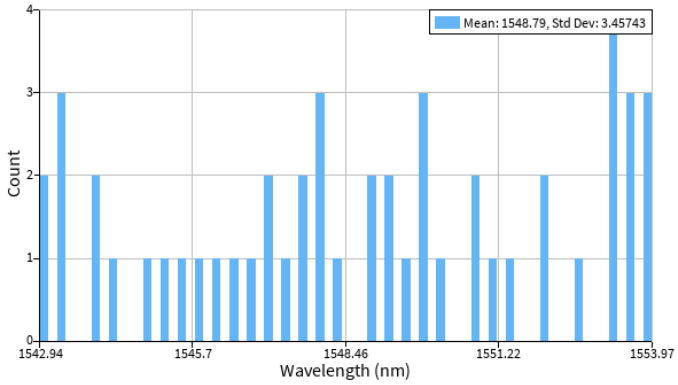


Fig. 9. Histogram of extracted center wavelength from the Monte Carlo simulation of the 50 µm MZI design. The simulated mean wavelength is 1548.79 nm with a standard deviation of 3.457 43 nm.

circuits were laid out using the course process design kit (PDK) using KLayout tool.

The fabricated chip contained the following MZI variants:

- MZI_TE_W500_DL30
- MZI_TE_W500_DL50
- MZI_TE_W500_DL100

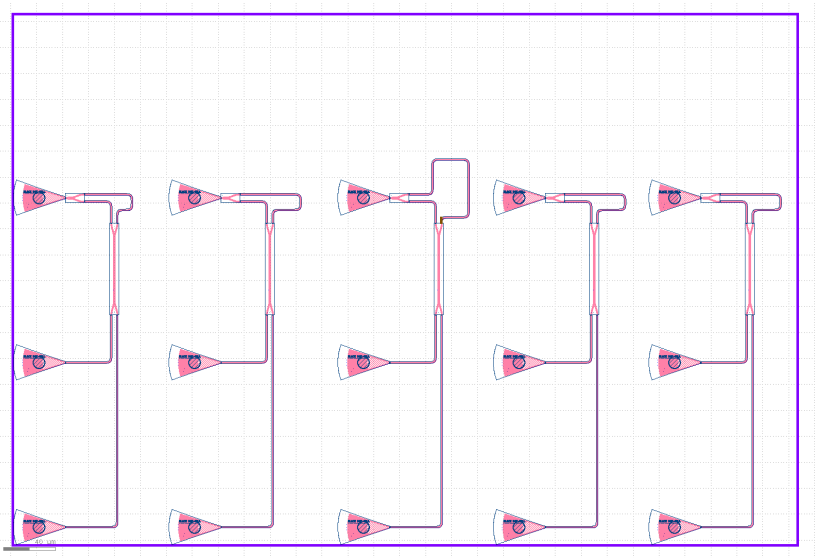

Fig. 10. Layout of the fabricated silicon photonics test structures, showing the five Mach–Zehnder interferometer variants.

- MZI_TE_W500_DL50_A
- MZI_TE_W500_DL50_B

Each device was accessed through TE grating couplers and measured over the wavelength range from $1.50\,\mu\mathrm{m}$ to $1.58\,\mu\mathrm{m}$. The fabricated circuits were designed to enable extraction of the waveguide group index from the measured free spectral range (FSR) of the MZI fringes and to compare the extracted values against the simulated compact waveguide model.

Figure 10 shows the device layout used for fabrication, including the grating couplers, Y-branch splitters/combiners, routing waveguides, and the five interferometer structures. The duplicated $50\,\mu\mathrm{m}$ designs were placed separately on the layout to allow comparison of process-induced variation across nominally identical devices.

## IV. Experimental Data

The devices were fabricated on a silicon-on-insulator platform with a $220\,\mathrm{nm}$ silicon device layer and oxide cladding using the course multi-project wafer flow. Depending on the foundry run, fabrication was carried out through either the Applied Nanotools NanoSOI process or the Washington Nanofabrication Facility (WNF) silicon photonics process. In both cases, the devices were patterned using 100 keV electron beam lithography[7] on HSQ resist, followed by anisotropic silicon etching and oxide cladding deposition. The fabrication flow included waveguide patterning, development, dry etching, and PECVD oxide deposition to complete the photonic structures. The final chip contained the five MZI variants used in this work: MZI_TE_W500_DL30, MZI_TE_W500_DL50, MZI_TE_W500_DL100, MZI_TE_W500_DL50_A, and MZI_TE_W500_DL50_B. Fabrication details were provided by the course infrastructure and foundry documentation. [13]

To characterize the devices, a custom-built automated test setup [8, 12] with automated control software written in Python was used [9]. An Agilent 81600B tunable laser was used as the input source and Agilent 81635A optical power sensors as the output detectors. The wavelength was swept from 1500 to 1600 nm in 10 pm steps. A polarization maintaining (PM) fibre was used to maintain the polarization state of the light, to couple the TE polarization into the grating couplers [10]. A 90º rotation was used to inject light into the TM grating couplers [10]. A polarization maintaining fibre array was used to couple light in/out of the chip [11].

The fabricated chip was characterized over the wavelength range from $1.50\,\mu\mathrm{m}$ to $1.58\,\mu\mathrm{m}$. For each MZI, the transmission spectrum was measured and the free spectral range (FSR) was extracted from the wavelength spacing between adjacent transmission maxima or minima near $1550\,\mathrm{nm}$. The Matlab code was used to plot the measurment data and curve fit to MZI transfer function and exact the group index for al five MZI designs.

Representative measured spectra for the three devices (dL of 30, 50 and 100um) are shown in Figs. 11–13. The extracted FSR and group index values are summarized in Table II.

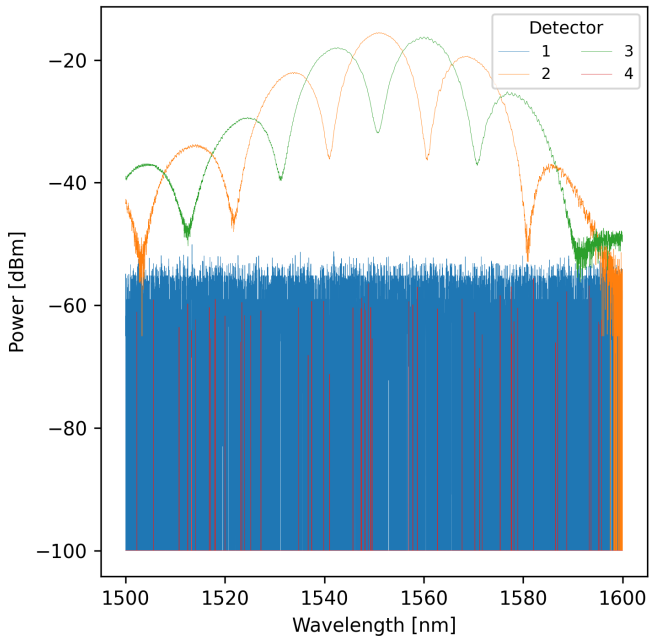


Fig. 11. Measured transmission spectrum of MZI_TE_W500_DL30.

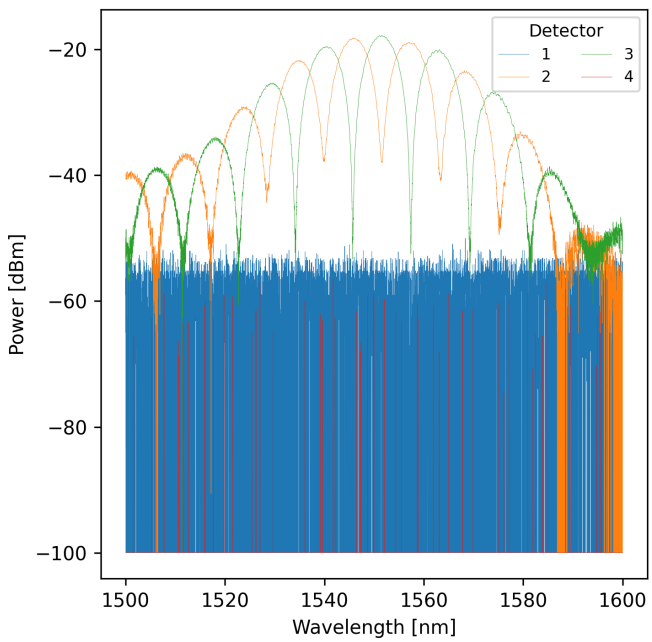


Fig. 12. Measured transmission spectrum of MZI_TE_W500_DL50.

## V. Analysis

The measured transmission spectra show the expected periodic interference fringes of unbalanced Mach–Zehnder interferometers. As predicted by the analytical model, the free spectral range decreases as the path length difference $\Delta L$ increases. The $30\,\mu\mathrm{m}$ device exhibits the largest measured FSR of $19.68\,\mathrm{nm}$, the $50\,\mu\mathrm{m}$ devices show FSR values around $11.5\,\mathrm{nm}$, and the $100\,\mu\mathrm{m}$ device exhibits the smallest FSR of $5.75\,\mathrm{nm}$. This trend is consistent with the inverse dependence of FSR on $\Delta L$ given by (14).

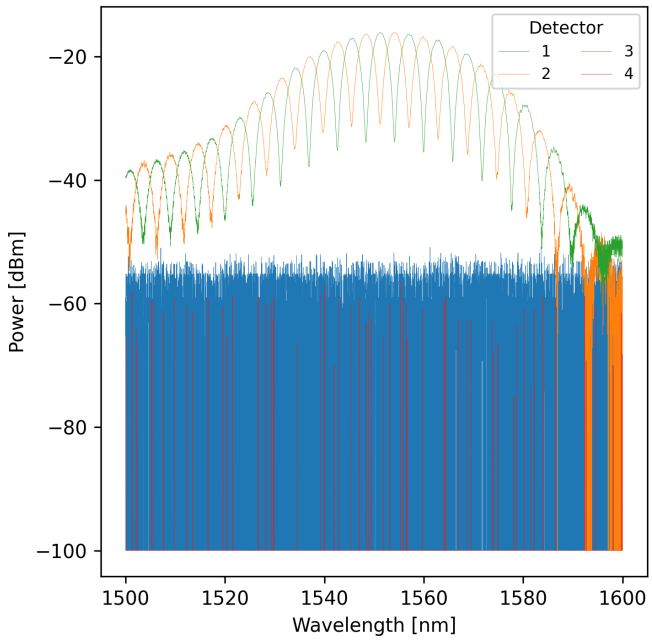


Fig. 13. Measured transmission spectrum of MZI_TE_W500_DL100.

TABLE II
MEASURED FSR AND EXTRACTED GROUP INDEX FOR FABRICATED MZI DEVICES.

| Design ID | $\Delta L$ ($\mu$m) | Measured FSR (nm) | Extracted $n_g$ |
|---|---|---|---|
| MZI_TE_W500_DL30 | 30 | 19.68 | 4.1166 |
| MZI_TE_W500_DL50 | 50 | 11.57 | 4.1259 |
| MZI_TE_W500_DL100 | 100 | 5.75 | 4.1576 |
| MZI_TE_W500_DL50_A | 50 | 11.60 | 4.1189 |
| MZI_TE_W500_DL50_B | 50 | 11.488 | 4.1103 |

TABLE III
COMPARISON OF SIMULATED AND MEASURED GROUP INDEX.

| Design ID | Calculated $n_g$ | Measured $n_g$ | Error (%) |
|---|---|---|---|
| MZI_TE_W500_DL30 | 4.197 | 4.1166 | -1.92 |
| MZI_TE_W500_DL50 | 4.197 | 4.1259 | -1.69 |
| MZI_TE_W500_DL100 | 4.197 | 4.1576 | -0.94 |
| MZI_TE_W500_DL50_A | 4.197 | 4.1189 | -1.86 |
| MZI_TE_W500_DL50_B | 4.197 | 4.1103 | -2.07 |

Using the measured FSR values, the group index was extracted for each device and compared against the simulated value of $n_g(1550\ \mathrm{nm}) = 4.197$. The comparison is summarized in Table III. All measured values are within approximately 2% of the simulated result, indicating good agreement between the compact waveguide model, the MZI transfer-function analysis, and the fabricated devices.

Table IV shows that the extracted FSR values agree well with the calculated values based on the simulated group index. The error is below 3.2% for all devices and below 1.5% for four out of the five devices, indicating that the fabricated MZIs closely follow the expected spectral behaviour.

Among the five devices, the $100\,\mu\mathrm{m}$ MZI produced the closest agreement with simulation, with an extracted group index of 4.1576. The two duplicate $50\,\mu\mathrm{m}$ devices, MZI_TE_W500_DL50_A and MZI_TE_W500_DL50_B, gave extracted group indices of 4.1189 and 4.1103, respectively, corresponding to a difference of only 0.0086. This

TABLE IV
COMPARISON OF CALCULATED AND EXTRACTED FSR VALUES FOR THE FABRICATED MZI DEVICES.

| Design ID | $\Delta L$ ($\mu$m) | Calculated FSR (nm) | Extracted FSR (nm) | Error (%) |
|---|---|---|---|---|
| MZI_TE_W500_DL30 | 30 | 19.08 | 19.68 | 3.14 |
| MZI_TE_W500_DL50 | 50 | 11.45 | 11.57 | 1.05 |
| MZI_TE_W500_DL100 | 100 | 5.72 | 5.75 | 0.52 |
| MZI_TE_W500_DL50_A | 50 | 11.45 | 11.60 | 1.31 |
| MZI_TE_W500_DL50_B | 50 | 11.45 | 11.488 | 0.33 |

TABLE V
COMPARISON OF MEASURED $50\,\mu\mathrm{m}$ MZI DEVICES WITH MONTE CARLO SIMULATION OF THE NOMINAL $50\,\mu\mathrm{m}$ DESIGN.

| Device | Measured FSR (nm) | MC Mean FSR (nm) | Deviation (nm) | Deviation / $\sigma$ |
|---|---|---|---|---|
| MZI_TE_W500_DL50 | 11.57 | 11.5088 | 0.0612 | 0.22 |
| MZI_TE_W500_DL50_A | 11.60 | 11.5088 | 0.0912 | 0.32 |
| MZI_TE_W500_DL50_B | 11.488 | 11.5088 | -0.0208 | -0.07 |

small difference indicates good fabrication repeatability and supports the robustness of the FSR-based extraction method.

The fit models extracted from the measured data also show broadly consistent behaviour across the different devices, with only modest variation in the fitted coefficients. These differences likely arise from a combination of fabrication-induced and measurement uncertainties. The overall agreement between simulation and experiment confirms that the compact waveguide model and the MZI-based group-index extraction method are valid for the fabricated structures.

The three nominally identical $50\,\mu\mathrm{m}$ imbalance devices, MZI_TE_W500_DL50, MZI_TE_W500_DL50_A, and MZI_TE_W500_DL50_B, were used to evaluate fabrication repeatability and to compare the measured spread against the Monte Carlo simulation of the nominal $50\,\mu\mathrm{m}$ design. Their measured FSR values were 11.57, 11.60, and 11.488 nm, respectively, corresponding to a mean measured FSR of 11.553 nm and a total spread of 0.112 nm.

The Monte Carlo simulation of the nominal $50\,\mu\mathrm{m}$ MZI design produced a mean FSR of 11.5088 nm with a standard deviation of 0.282958 nm. Table V compares the measured FSR values of the three fabricated devices against this simulated distribution. All three measured devices lie very close to the Monte Carlo mean, with deviations of only 0.0612 nm, 0.0912 nm, and $-0.0208$ nm, respectively. Expressed in units of the simulated standard deviation, these correspond to approximately $0.22\sigma$, $0.32\sigma$, and $-0.07\sigma$, respectively. These values indicate that all three measured devices fall well within the expected process-variation window predicted by simulation. The corresponding relative deviations from the Monte Carlo mean are approximately $0.53\%$, $0.79\%$, and $-0.18\%$, respectively.

This agreement between measured and Monte Carlo simulated results is important for two reasons. First, it confirms that the repeated $50\,\mu\mathrm{m}$ devices are consistent with normal fabrication-induced variation rather than showing any anomalous behaviour. Second, it supports the validity of the compact waveguide model and the process assumptions used in the Ansys numerical simulation. In other words, the fabricated duplicated devices not only agree with one another, but also fall within the statistically expected spread of the simulated design.

## VI. CONCLUSION

The Mach-Zehnder Interferometer designs were proposed, simulated, and experimental characterization were carried out for multiple silicon strip waveguide MZI implementations with different path length differences. The measured spectra showed the expected interference behaviour, and the extracted FSR values followed the same trend as the analytical and simulated results.

The experimentally extracted group indices were all within about 2% of the simulated value, confirming that the proposed MZI-based method provides a reliable way to extract the waveguide group index from fabricated devices. In addition, Monte Carlo simulation of the repeated $50\,\mu\mathrm{m}$ MZI design showed that the measured duplicated devices fall well within the simulated process-variation distribution.

## VII. Acknowledgment

I acknowledge the edX UBCx Phot1x Silicon Photonics Design, Fabrication and Data Analysis course, which is supported by the Natural Sciences and Engineering Research Council of Canada (NSERC) Silicon Electronic-Photonic Integrated Circuits (SiEPIC) Program. The devices were fabricated by Richard Bojko at the University of Washington Washington Nanofabrication Facility, part of the National Science Foundation's National Nanotechnology Infrastructure Network (NNIN), and Cameron Horvath at Applied Nanotools, Inc. Mateo Branion-Calles performed the measurements at The University of British Columbia. I acknowledge Lumerical Solutions, Inc., Mathworks, Mentor Graphics, Python, and KLayout for the design software.

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